

Are the Skilled Trades Safe From AI?

The planning guide — the full scores by track, the six factors behind them, the honest trade-offs, and what to ask before you commit.

2.5

AI EXPOSURE · ELECTRICIAN — SERVICE & MAINTENANCE
where 10 is most at risk

For anyone still choosing — read it straight through.



Before you read this

You're weighing the skilled trades — or a path into a trade — and you want to know what that choice actually looks like in an AI economy. That's the whole job of this guide.

It's written to you directly, whether you are the one choosing or a parent thinking it through alongside them. Where it says "you," a parent can read "them"; it's the same decision from either side.

What this is. A facts document, not a recommendation. It sets out what's known, where the evidence is contested, and where we might be wrong — and then how to choose well given all of it.

If you're the parent, a note on how to use it. Don't lead with the scores. Open a conversation with a risk number and one of two things happens: they shut down, or they get defensive about a choice they're already leaning toward. Read it yourself first, then share it as information you both react to, not a verdict you've reached. And be honest with yourself about your own first reaction, because for many parents this is the choice they least expected to be reading about — if your instinct was disappointment, that's worth examining before the conversation rather than during it. The scores here are better than most degrees, the debt position is better than all of them, and the demand picture is stronger than anything else we measure. The disappointment is usually about status rather than outcomes.

And one thing specific to the trades — honest on both sides: this is not a consolation prize, and it is also not for everyone. The physical demands over thirty years are real, the income ceiling without business ownership is real, and someone who dislikes being outdoors in February will dislike it for a very long time. Good news should open the conversation, not close it.

The short answer

For the right person, this is the strongest bet on the board — and almost nobody considers it seriously.

A service electrician scores 2.5 out of 10 — among the lowest scores in this entire index. No tuition debt. An on-ramp that is widening while most degree routes narrow. And the demand

picture is the strongest we measure anywhere.

The protection was never "working with your hands." It was working with your hands somewhere unpredictable. That distinction is the whole guide, and it explains why the same skills on a production line score twice as high — which is exactly why which version you choose still matters, even in the most protected career we measure.

The scores

TRADE AND SETTING	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Electrician — service & maintenance	2.1	2.3	2.5	+0.4	Very low
Plumbing / HVAC — service	2.2	2.4	2.6	+0.4	Very low
Welding & fabrication — site work	2.8	3.0	3.2	+0.4	Low
Construction site trades	2.8	3.1	3.4	+0.6	Low
Production line / prefab / modular	4.0	4.5	5.0	+1.0	Moderate

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, bedside nursing scores 2.8, licensed engineering 4.0, a business analyst 7.6, and entry-level software development 8.1.

Read the table this way: the service electrician's +0.4 over three years is the slowest movement in the entire index. Entry-level software moved +1.8 over the same period. And note the last row — the same trade skills in a factory setting move at more than twice the rate of the same skills in the field. Which setting you aim at is a real choice, even here.

Why the trades score so low — the six factors



Here's how a service electrician answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

The paperwork, and essentially nothing else.

Quoting and estimating. Scheduling and dispatch. Invoicing and job admin. Code and regulation lookup. Parts ordering and inventory.

For a self-employed tradesperson, almost all of that is a benefit rather than a threat. The office burden of running a trade business is the part being automated.

What resists: diagnosing a fault in a sixty-year-old building, running new service through a wall with no drawings, working in an attic or a crawlspace or a trench, and judging whether conditions are safe to continue.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Among the best on-ramps we measure anywhere, and it is getting better rather than worse.

Apprenticeships are paid rather than charged for, legally embedded in licensing, and chronically undersubscribed. The route in is widening at exactly the moment most degree routes are narrowing.

BLS projects **649,300 annual job openings** in construction and extraction occupations over 2024–2034. Electricians specifically show **9% growth** — classified as "much faster than average" — with roughly **81,000 openings each year** against total employment of about 818,700.

One fork worth seeing even here, where entry is easy. The setting you enter shapes the protection you inherit. The same trade skills on a production line or in a prefab facility score 5.0

— double the field electrician's 2.5 — because that is the controlled environment automation handles best. The way through is to enter on the pathway that leads to a license and breadth of field work — a registered or union apprenticeship in service, maintenance and retrofit — rather than whichever placement is hiring fastest. An easy start in repetitive new-build or line work is not the same as landing in the most protected version of the trade.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Completely, and in the hardest possible setting for automation.

This is the factor that does most of the work, and the section on why "manual work is safe" is the wrong idea explains why "physical" is not one thing.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Yes, and more than people assume.

Someone is being let into a home. Someone is signing off that the work is safe. When an installation fails and causes a fire or a flood, a licensed human is answerable.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

For electrical and gas work, comprehensively.

Licensing, inspection and certification requirements are written into building codes and enforced by inspectors. Nobody signs off their own unlicensed electrical work.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Every single job.

A different building, unlabeled wiring, work someone else did badly in 1974, and a customer describing the problem inaccurately. The environment is different every time, which is precisely the condition robotics handles worst.

"Manual work is safe" is the wrong idea

This deserves its own section, because the popular version of this argument is wrong in a way that matters — and the trades are where it breaks most visibly.

The usual framing is binary: desk jobs are exposed, manual jobs are safe.

What actually matters is not whether the work is physical. It's whether the environment is predictable.

Robotics performs well in controlled settings — known dimensions, repeated tasks, fixed tolerances, no surprises. That is a warehouse, an assembly line, a modular construction facility. It performs badly in uncontrolled ones, and that gap is not closing quickly.

A service electrician spends the day in unpredictable environments by definition. That rates **9.5** on our physical factor.

The same trade skills applied to a production line rate **8.0** — still physical, but in exactly the setting robotics handles best. That single difference, plus weaker licensing and lower trust requirements, is what separates a 2.5 from a 5.0.

So the useful question for any hands-on career is not "is this manual work?" but "would two days of this job ever look the same?" If the answer is no, the protection is real. If the answer is yes, physical work offers much less shelter than people assume.

This is also the thing to watch over a career. Modular and offsite construction is growing, and it works by moving site work into factories. It is the one development that could genuinely move these scores — not because robots got better at crawlspaces, but because the industry moved the work somewhere robots already function.

What the trend shows

Service electrician: 2.1 → 2.3 → 2.5. Up 0.4 in three years — tied for the slowest movement in the entire index, and all of it administrative.

Production line / prefab: 4.0 → 4.5 → 5.0. Up 1.0, more than twice the pace.

The gap widened from 1.9 points to 2.5. The pattern is now familiar: the unpredictable end holds while the structured end drifts. What is unusual here is how little the protected end has moved at all.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Strongest in the index	The one to watch	Not from robots reaching crawlspaces, but from prefabrication moving work into factories.
Regulatory / licensing	Strong for electrical and gas	Most durable	Building codes and inspection regimes move at the speed of legislation.
Judgment under novelty	Very strong	Durable	Every building is different. No two service calls are alike.
Human trust / accountability	Strong	Durable	Someone is in a home, and someone signs off on safety.

The honest read for someone at the start. You'll work for roughly forty-five years — longer than most graduates, because you start earlier. Over that span, the licensing and novelty protections should hold comfortably. The physical protection is the strongest today and the one with the most credible long-term threat — and the threat is industrial reorganisation rather than robotics.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Working out the quote by hand	Software quotes it; you check it's right
Ringling round for parts	Ordering is automated
Paperwork in the evening	Largely gone
Diagnosing from experience	Diagnosing from experience, plus better instruments
Learning codes by memory	Looking them up instantly on site
The actual work	Unchanged

The last row is the point. In most professions the automatable layer and the skilled layer are tangled together. In the trades they are cleanly separable, and what's automating is the part nobody enjoys.

Human strengths that will matter most

1. **Diagnosis in unfamiliar conditions** — working out what's wrong in a building nobody documented. The core skill and the least automatable thing here. 2. **Physical judgment** — knowing whether something is safe to proceed with, by feel and by eye. 3. **Being trusted in someone's home** — a genuine commercial asset, and the foundation of a business. 4. **Customer handling** — explaining a problem plainly to someone worried about the cost. 5. **Business sense** — the thing that separates a \$60,000 tradesperson from a \$200,000 one.

And underneath all five, two traits the score can't grade: **curiosity and adaptability.** The specific tools — the diagnostic instruments, the thermal camera, the quoting and scheduling software — will turn over many times across a career, and so will the specializations that pay best. The tradespeople who compound are the ones who stay genuinely curious and re-tool fastest — moving toward data centers, EV infrastructure, solar and heat pumps as the demand shifts — rather than defending the one way of working they trained on. In a trade where the ceiling is reached by where you take the skill, that flexibility is itself a form of protection.

Other things to consider

What's genuinely good about it

The financial position is better than most families realize, and the comparison usually never gets made.

- **No tuition debt.** Apprentices are paid from day one — typically starting at \$15–20 an hour and stepping up annually.
- **Four to five years to journeyman status,** earning throughout.
- **BLS median wages:** electricians **\$62,350**, plumbers **\$62,970**, with top earners in both exceeding **\$106,000**.
- **Union total compensation runs roughly 60% higher** than non-union once health, pension and benefits are counted, per BLS.

- **Construction wages grew 4.2% year-over-year** into mid-2025, outpacing the national average across all occupations.

The comparison worth running: a graduate finishes at 21 or 22 with debt and enters a market where 41.5% of recent graduates are underemployed. An apprentice finishes at 22 or 23 with no debt, four years of earnings behind them, a license, and — currently — employers competing for them.

And what tradespeople report as rewarding is consistent and distinctive: the problem is solved and visible by the end of the day, the work is genuinely varied, autonomy is high, and running your own business is a normal outcome rather than an exotic one.

What's genuinely hard about it

The physical toll over decades is the real cost, and it is the thing most likely to end a career early. Knees, backs and shoulders. The work is done in heat, cold, confined spaces and awkward positions.

To be fair to the data: the nonfatal injury rate for specialty trade contractors is **2.3 per 100 full-time workers** and has been declining, driven by sustained focus on falls, struck-by, electrocution and caught-in hazards. It is safer than it was. It is still harder on a body than any desk job, and that compounds over thirty years.

The income ceiling without ownership is real. Median pay is solid and the top end is excellent, but the path to the top end usually runs through running a business — which is a different skill from the trade itself, and not everyone wants it.

It is cyclical. Construction rises and falls with credit conditions and public spending in a way that nursing and teaching do not.

And the status question is real, even if it shouldn't be. Some people find the condescension wearing, particularly from people who assume it was a fallback.

Who this work suits — and who it doesn't

The trades tend to suit people who:

- **Want the problem solved by the end of the day.** The single most-cited satisfaction.
- **Would rather be moving than sitting**
- **Like diagnosis** — the puzzle of working out what's actually wrong
- **Are practical with people** — most trades involve customers, not just materials

- **Are interested in eventually running something**

It's harder for people who:

- Genuinely **dislike physical discomfort**. Being outdoors in February is not a phase.
- Want **predictable indoor conditions**
- Need **structured career progression** with titles and ladders
- Are **conflict-averse about money** — chasing invoices is part of self-employment

What else is moving this market

The shortage is severe and well documented. The construction industry needs to attract hundreds of thousands of net new workers in 2026 alone, against an ageing existing workforce.

And here is the detail that reframes this entire guide. Microsoft's president Brad Smith has publicly identified the electrician shortage as the single biggest constraint on data center expansion — not chip supply, not land, not permitting. Industry estimates put the requirement at 300,000-plus additional electricians just to meet AI-related data center demand. Electricians working on data center projects report earnings well into six figures.

Capital has noticed: BlackRock launched a \$100 million initiative in March 2026 aimed at training 50,000 skilled trade workers, and the US Department of Labor announced \$145 million in apprenticeship grants targeting shipbuilding, defense, semiconductors and energy.

So the AI buildout is simultaneously the thing threatening white-collar entry work and the thing driving demand for electricians. That is not irony — it is the most concrete illustration in this index of where the protection actually lies.

One honest caution. A meaningful share of current construction demand is project-driven — data centers, infrastructure programs — and those end when the builds do. The demographic case, an ageing workforce retiring faster than replacements arrive, holds regardless of the cycle. **Weight the demographics more heavily than the boom.**

The AI-native advantage — what to actually do about it

The trades are unusual here: AI helps a tradesperson mainly by removing the parts of the job that aren't the trade.

1. Run the business side properly. Quoting, scheduling, invoicing and customer follow-up are now largely automatable. **The tradespeople who use this well aren't better at the trade — they're**

better at running a trade business, which is where the income ceiling actually sits.

2. Use diagnostic tools without deferring to them. Thermal imaging, circuit analysis and fault-finding tools are improving. They inform judgment; they don't replace it.

3. Look things up rather than memorise them. Codes change constantly. Instant lookup is a genuine efficiency gain.

4. Specialize deliberately. Data centers, EV infrastructure, solar and battery, heat pumps. These are where the premium is and where demand is growing fastest.

5. Get to ownership. Everything in this guide points at it. The trade gives you a protected skill; the business is what converts it into a ceiling.

Concrete preparation: get any hands-on experience before committing — a weekend with someone working is worth more than any brochure. Compare apprenticeship providers on completion rates and on what happens after qualification. And choose the trade deliberately: electrical leads on pay, growth and protection, and it is the one sitting directly in front of the AI-driven demand wave.

The routes in — and the versions that hold up

ROUTE	NOTES
Registered apprenticeship	The main route. Paid from day one, 4–5 years to journeyman, nationally recognized credential.
Union apprenticeship	Structured, well-paid, with substantially higher total compensation. Competitive entry.
Pre-apprenticeship program	For someone with no experience. Short, and a route into the above.
Trade school then apprenticeship	Faster classroom start, but costs money the apprenticeship route doesn't.
Military technical training	Excellent trade grounding with civilian licensure pathways.

The thing to compare: completion rates and post-qualification placement, not marketing. And the version that holds up is the one that leads to a license and to breadth of field work — service, maintenance and retrofit — rather than narrow or repetitive line work, which sits at the 5.0 end of the table.

Where this can lead later

The ceiling is higher than most people assume, and it is reached differently.

- **Journeyman → master → contractor** with your own license
- **Business ownership** — the main route to the top end
- **Site supervision and construction management** — trade-to-supervision remains a fully respected path, and construction management scores 3.3
- **Specialist and high-demand work** — data centers, industrial, marine, controls
- **Inspection, code enforcement and safety**
- **Teaching the trade** — instructor roles at colleges and apprenticeship providers
- **Estimating and project management** — note this moves toward the desk, and the score climbs accordingly

The pattern holds here as everywhere: moving away from the physical work raises exposure. Estimating scores 6.3. It's a good career and a different one.

What to look for in an apprenticeship

1. **Completion rate.** The number that matters most and the one least advertised.
 2. **Breadth of work.** Domestic only, or commercial and industrial too? Breadth determines the ceiling.
 3. **What happens at the end.** Is there a job, and at what rate?
 4. **Licensing pathway.** Does the program lead cleanly to the license, or leave gaps?
 5. **Quality of supervision.** In an apprenticeship, the person teaching you is the education.
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Questions to ask an apprenticeship provider or employer

- What proportion of apprentices complete, and what happens to those who don't?

- What's the pay progression year by year, and what do journeymen here earn?
- What range of work will I see — domestic, commercial, industrial?
- Do you have a job at the end, and what proportion of apprentices stay on?
- What license does this lead to, and is anything else required?
- Who supervises apprentices day to day, and how many do they carry at once?

Red flags: vagueness about completion rates; narrow work only; no clear licensing pathway; apprentices used as unsupervised labor.

The career-hunting checklist

Choosing well is step one. This is what turns a good choice into a good outcome once you're in — worth reading now, because the best time to pick a trade and a provider is knowing exactly what you'll need to do with it. The first list is for the apprenticeship years; the second is for the job hunt that follows. In the most protected career on the board, the work here is less about escaping exposure and more about aiming at the most durable version of the trade.

During the apprenticeship — build toward the license and the ceiling

- **Choose the trade deliberately, and lean electrical.** Electrical leads on pay, growth and protection (service electrician 2.5), and it sits directly in front of the AI-driven data center demand wave. Plumbing/HVAC service (2.6) is close behind. Whichever you pick, pick it on purpose.
- **Aim at the field, not the factory.** Service, maintenance and retrofit work is the protected end (2.5–2.6); production-line, prefab and modular work scores double (5.0) because it's the controlled setting automation handles best. Enter where the environment is different every time.
- **Treat the license as the product, not the placement.** Confirm the program leads cleanly to the license and to breadth of work — domestic, commercial and industrial — because breadth determines the ceiling and the license is the legal protection.
- **Get real hands-on exposure before you commit.** A weekend with someone working, including the unglamorous parts and February weather, tells you more than any brochure — and confirms the physical side is one you can live with for decades.
- **Get fluent with the business and diagnostic tools as a professional, not a shortcut.** Learn the quoting, scheduling and invoicing software that's automating the office

burden, and use thermal imaging and fault-finding instruments to inform judgment rather than replace it.

- **Feed curiosity and adaptability on purpose, and plan for ownership from the start.** The premium specializations — data centers, EV, solar and battery, heat pumps — shift over a career, and the income ceiling runs through running a business. The tradesperson who re-tools fastest and builds toward ownership compounds.

During the job hunt — aim for the durable work, don't just apply

- **Target the protected lanes on purpose.** Service, maintenance and retrofit field work, ideally electrical, ideally in high-demand specialisms — not whichever placement appears first.
- **Read the setting, not just the offer.** Repetitive production-line, prefab and modular work is the 5.0 rung — the controlled setting automation moves fastest into. If you take one, take it as a deliberate stepping-stone with an exit in mind, not as the plan.
- **Don't grab a role just because it's on offer in an exposed lane.** A role being available doesn't make it safe. An employer still running a large repetitive line or prefab intake may simply be behind the curve — and that's exactly the setting the score says drifts fastest. Availability can be a lagging indicator, not a green light.
- **Prefer breadth and the license.** A seat that gives you commercial and industrial work and counts toward the license beats a narrow domestic-only role that caps the ceiling, even if the second pays a little more today.
- **Ask the employer where AI and prefabrication are heading in the work.** Whether they're moving toward offsite modular assembly or staying in field service and retrofit tells you whether the seat holds its protection or quietly drifts toward the 5.0 end.
- **Weigh the trajectory and the body over the starting wage.** A role that builds toward ownership, breadth and a high-demand specialism beats a higher-paid seat in repetitive work that's harder on the body and closer to the automating end.

Go deeper — further reading

- **Electricians — Occupational Outlook Handbook** — US Bureau of Labor Statistics. Primary source and free: 9% projected growth through 2034, roughly 81,000 annual openings, wage distributions and entry requirements. Look up plumbing and HVAC while you're there.

- [Electrician career guide 2026](#) — the data center story in detail, including Microsoft's identification of the electrician shortage as the primary constraint on expansion, and the industry estimate of 300,000-plus additional electricians needed.
- [Skilled trades statistics for 2026](#) — wages, injury rates, union compensation differentials and the BLS opening projections collected in one place.

The counterargument — read this one:

The strongest case against our score isn't that robots will reach crawlspaces. It's **prefabrication**.

Modular and offsite construction is growing steadily, and it works precisely by moving work from unpredictable sites into controlled factories — the setting where our own framework says automation performs best. On this reading, the trades' protection erodes not because the work becomes automatable, but because the industry relocates it somewhere it already is.

Our own table concedes the mechanism: production-line and prefab work scores 5.0 against 2.5 for field service. If the share of construction done offsite rises substantially, the average tradesperson's exposure rises with it — and our headline figure would be describing a shrinking share of the trade.

Where we land: we hold the score because service, maintenance and retrofit work — which is a large and growing share of trade employment, driven by an ageing building stock — cannot be prefabricated. You cannot factory-build a repair to a 1963 house. But we'd treat rising offsite construction share as the single most important thing to watch in this guide, and we'll report it if it moves.

Bottom line

For the right person this is the strongest position in the index, and it is chronically under-recommended to exactly the people who would thrive in it — which is the good news for someone still choosing: the decisive choices, of trade and of setting, are the ones in front of you.

The scores are among the lowest we measure. There is no tuition debt. The on-ramp is widening while most degree routes narrow. Demand is severe, well-documented, and — in electrical work — being driven by the AI buildout itself.

The protection is specific and worth understanding rather than assuming. It is not that the work is manual. It is that **the environment is different every time**, which is the condition automation handles worst. Move the same skills into a factory and the score doubles.

The honest costs are real and have nothing to do with technology. Physical wear over decades is the thing most likely to end a career early. The income ceiling without business ownership is genuine. And construction is cyclical in a way that healthcare and education are not.

So the useful advice is: choose the trade deliberately rather than taking whichever apprenticeship appears first, aim at service and retrofit work rather than repetitive new-build, compare providers on completion rates, and plan for ownership from the start rather than discovering at 35 that the ceiling is lower than expected.

And if your first reaction to this was disappointment — it's worth sitting with where that came from. On the evidence in front of us, this is a better financial and occupational position than most degrees currently offer.

Questions worth sitting with

Whether you're the one choosing or a parent thinking it through together, these are the conversations that lead to a better decision. Read "you" as whoever the choice is about.

About the work itself

1. What made the trades appealing? Push past "I'm practical" to something specific. 2. Do you like *diagnosis* — working out what's wrong — or *building*? Different trades reward each differently. 3. Have you done anything hands-on for a sustained period? What did you think? 4. How do you actually feel about being cold, wet or in a confined space? Not in theory — in practice.

Reacting to the findings

5. A service electrician scores 2.5 and an entry-level software developer 8.1. Was that surprising? 6. Production line work scores twice as high as field service. Does that change which trades or settings appeal? 7. The guide argues the protection is unpredictability rather than manual work. Does that land?

The harder conversation

- 8. **The physical toll over thirty years.** Have you thought about what that actually means at 50?
- 9. The income ceiling usually runs through owning a business. Does that appeal, or is it a chore?
- 10. Construction is cyclical. What's the plan in a downturn?

The money question

- 11. Have you actually compared the numbers? An apprentice earning from 18 with no debt, against a graduate starting at 22 with debt in a market where 41.5% of graduates are underemployed.
- 12. Do you know union apprenticeships carry roughly 60% higher total compensation? Have you looked at what's available locally?

Testing it against reality

- 13. **The most valuable single action:** spend a day with someone working. Not a college open day — an actual working day, including the unglamorous parts. Then find someone ten years in and ask what their body feels like, and whether they'd do it again.

Widening the frame

- 14. Which trade, and why? Electrical leads on pay, growth and protection — have you considered that specifically? Have you looked at construction management, where trade-to-supervision scores 3.3?
- 15. If the trades weren't available, what would you choose? The answer usually reveals what you're actually drawn to.

This is analysis and informed judgment about a fast-moving situation, not a guarantee or a prediction. For the trades specifically, we've flagged that the most credible long-term threat is prefabrication moving work into controlled settings, rather than robotics reaching uncontrolled ones.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Skilled Trades

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned} \text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment}) \end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

ELECTRICIAN — SERVICE & MAINTENANCE — 2.5 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.9	4.4	4.9	1.72
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	9.0	9.0	9.0	0.1
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	2.52 → 2.5

PLUMBING / HVAC — SERVICE — 2.6 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.1	4.5	5.0	1.75
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	8.8	8.8	8.8	0.12
Judgment under novelty	10%	8.8	8.8	8.8	0.12
				Total	2.59 → 2.6

WELDING & FABRICATION — SITE WORK — 3.2 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.5	5.0	5.4	1.89
Entry-path erosion	15%	2.0	2.2	2.5	0.38
Physical / embodied	15%	9.2	9.2	9.2	0.12
Human trust / accountability	15%	7.5	7.5	7.5	0.38
Regulatory / licensing	10%	7.5	7.5	7.5	0.25
Judgment under novelty	10%	8.2	8.2	8.2	0.18
				Total	3.19 → 3.2

CONSTRUCTION SITE TRADES — 3.4 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.3	5.0	5.7	1.99
Entry-path erosion	15%	2.0	2.4	2.8	0.42
Physical / embodied	15%	9.2	9.2	9.2	0.12
Human trust / accountability	15%	7.5	7.5	7.5	0.38
Regulatory / licensing	10%	7.0	7.0	7.0	0.3
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	3.41 → 3.4

PRODUCTION LINE / PREFAB / MODULAR — 5.0 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.9	5.1	6.4	2.24
Entry-path erosion	15%	3.5	4.0	4.5	0.67
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	5.0	5.0	5.0	0.75
Regulatory / licensing	10%	4.5	4.5	4.5	0.55
Judgment under novelty	10%	5.0	5.0	5.0	0.5
				Total	5.01 → 5.0

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Electrician — service & maintenance

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	3.9	1.5	0.5	2.1
2025	4.4	1.8	0.5	2.3
Fall 2026	4.9	2.0	0.5	2.5

Movement decomposition: automatability contributed **+0.35** and entry-path erosion **+0.07**, for a total of **+0.4** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.